2.1 INTRODUCTION

In this modern world, electricity plays a major role in human life. This is because, electrical appliances play an important role in our day to day life. According to theory, the current flow is nothing but flow of electrons. The force required to move electrons from higher level to lower level is called voltage.

Hence conductors are required to conduct the current from one place to another place. They are classified in to three types, namely solid conductors, liquid conductors and gaseous conductors. The material which does not conduct current is called an insulator. Another one is a semi-conductor, which has the property in between the conductor and insulator. Semi-conductors are used in making electronic devices.

First, let us study the types and properties of conductors.

2.2 CONDUCTORS

The wire which carries or conducts the current from the supply to the load is called conductors. The example of loads are fan, bulb, mixie, motors etc. Based on the state the conductors, they are classified into three types:

i) Solid Conductors
ii) Liquid Conductors
iii) Gas Conductors

i) Solid Conductors

These conductors are converted into thin wire, thin rod or strap for the purpose of conduction.

Some of the examples of solid good conductors are Silver, Copper, Brass, Aluminium, Tungsten, etc.

ii) Liquid Conductors

The conductors in the form of liquids are called liquid conductors. Liquid conductors are mostly used in batteries.

Example: Mercury, Sulphuric Acid, Nitrate etc.

Mercury is used in high power vapour lamp and automatic circuit breakers.

iii) Gas Conductors

The conductors in the form of gas are called gas conductors. These are used in gas discharge lamps at high temperature which are used in big shops, malls, etc.

2.2.1 Properties of Conductors

So far we have studied about the types of conductors. Let's see the properties of good conductors.

- They conduct the current easily, because they have low resistance
- Have high tensile stress
- They are more flexible
- They are not affected by corrosion due to air, rain, heat, etc
- They are not affected by heat produced in conductor, when current flow takes place

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